MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

OCTANT/QUADRANT, ANALOG, KEY CODE SHV

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic module.

* 1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.93 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be SHV.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/188-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage of 5.0 ±0.5 V dc and ±15.0 ±1.5 V dc.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

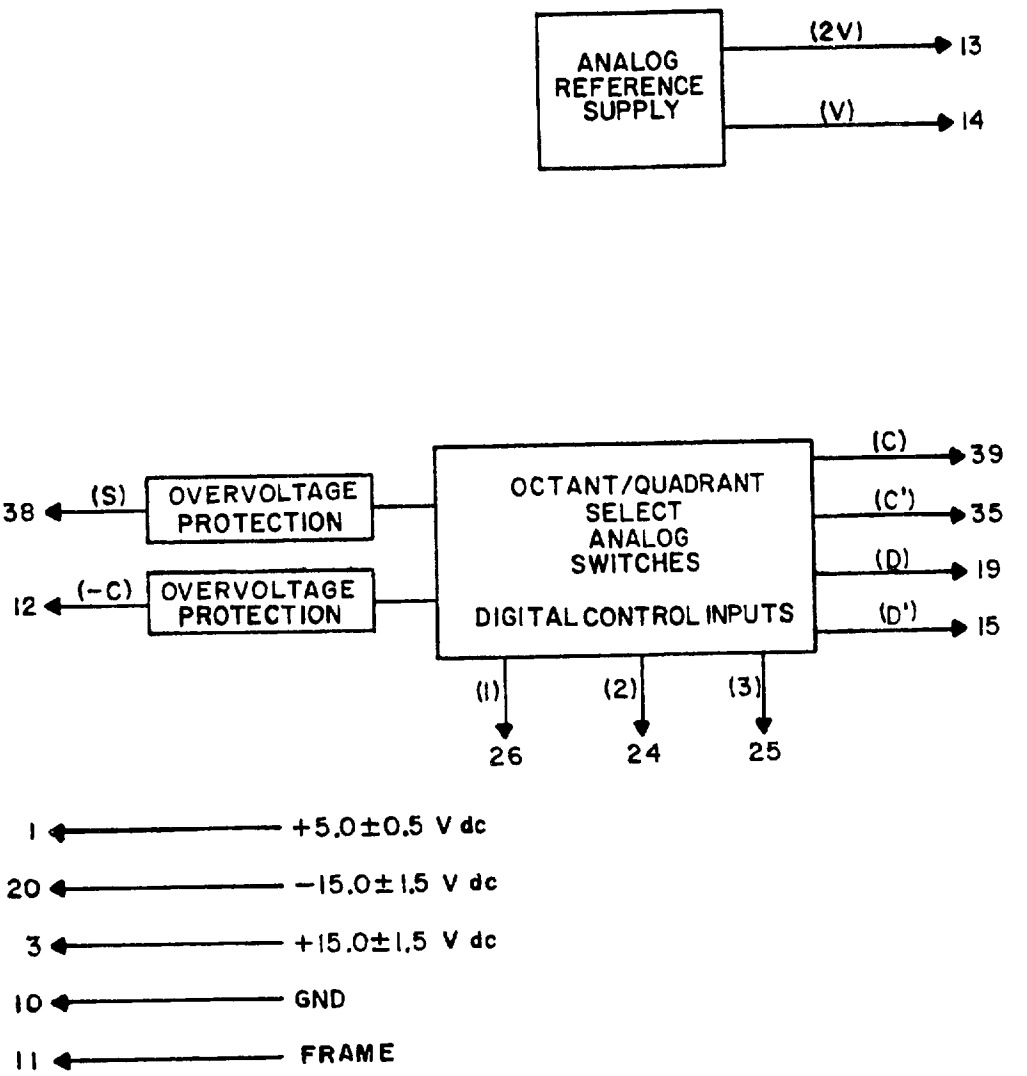
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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NOTE: The following are unused pins: 2, 4, 5, 6, 7, 8, 9, 16, 17, 18, 21, 22, 23, 27, 28, 29, 30, 31, 32, 33, 34, 36, 37, and 40.

* FIGURE 1. SHV module function diagram, octant/quadrant.

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*TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A25	Isolation	II	S	10	∞	1	∞	10	∞	1	∞	MΩ
B1,B2	Stress (overvoltage protection) check	III	Q	← Survival →								
C	Overvoltage protection check	IV	S	← Survival →								
D1-D24	Hot socket	V	Q	← See table V →								
E1-E3	Input breakdown current (digital)	VI	S	0	100	0	100	0	100	0	100	μA
F1-F4	Output short circuit current	VII	S	10	50	10	50	10	50	10	50	mA
G	V_S stress check	VIII	Q	← Survival →								
H1-H19 odd, H20-H27	Angular accuracy check 4/	IX	X	-0.030	0.030	-0.030	0.030	-0.030	0.030	-0.030	0.030	Degree
H2-H18 even	Quadrature	IX	X	-2	2	-2	2	-2	2	-2	2	mV
J1,J3	Absolute Tr	X	X	44.7121	45.2851	44.7121	45.2851	44.7121	45.2851	44.7121	45.2851	Degrees
J5	Absolute Tr	X	X	J1- 0.040	J1+ 0.001	J1- 0.040	J1+ 0.001	J1- 0.040	J1+ 0.001	J1- 0.040	J1+ 0.001	Degree
J6	Absolute Tr	X	X	J3- 0.040	J3+ 0.001	J3- 0.040	J3+ 0.001	J3- 0.040	J3+ 0.001	J3- 0.040	J3+ 0.001	Degree
J2,J4	Phase shift (quadrature)	X	X	-0.5	0.5	-0.5	0.5	-0.5	0.5	-0.5	0.5	Degree
K1	Power supply current (+5.5 V)	XI	S	0	7	0	7	0	7	0	7	mA
K2,K3	Power supply current (±16.5 V)	XI	S	0	20	0	20	0	20	0	20	mA
L1,L3	Reference supply output voltage	XII	S	0	10	0	10	0	10	0	10	mV rms
L2	Reference supply output voltage	XII	S	5.8	6.6	5.8	6.6	5.8	6.6	5.8	6.6	V
L4	Reference supply output voltage	XII	S	4.3	4.9	4.3	4.9	4.3	4.9	4.3	4.9	V

See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
M1	Reference supply output short circuit current	XII	S	1.5	25.0	1.5	25.0	1.5	25.0	1.5	25.0	mA
M2	Reference supply output short circuit current	XII	S	10	30	10	30	10	30	10	30	mA
N1-N4	Output offset voltage (analog)	XIII	X	-25	25	-25	25	-25	25	-25	25	mV
P1-P3	Input current (digital), high level	XIV	X	0	10	0	10	0	10	0	10	µA
P4-P6	Input current (digital), low level	XIV	X	-10	-200	-10	-200	-10	-200	-10	-200	µA
Q	Continuity, frame	XV	S	0	1	0	1	0	1	0	1	Ω
R1-R17 odd	Angular accuracy (800 Hz) check 4/	XVI	Q	-0.030	0.030	-0.030	0.030	-0.030	0.030	-0.030	0.030	Degree
R2-R18 even	Quadrature (800 Hz)	XVI	Q	-4	4	-4	4	-4	4	-4	4	mV
S1-S3	Input capacitance (digital)	XVII	Q	2	15	2	15	2	15	2	15	pF
T1-T16	Noise rejection	XVIII	S	0.0	3.5	0.0	3.5	0.0	3.5	0.0	3.5	mV rms
U1,U2	Input impedance (power off)	XIX	S	0	240	0	240	0	240	0	240	mVp

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Limits for test conditions H1-H19 odd, H20-H27, and R1-R17 odd are given as minus and plus values of the selected angle.

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4.4.2 Order of testing. Test conditions A through U shall be performed in any order with the following exceptions. If test conditions A,B,C,D,E,F, or G are part of a sequence of tests, the order of testing shall be as follows: A,B,C,D,E,F,G where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through U in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions H through R, T, and U shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests H through U shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests H through U shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

* 6.3 Cross reference of classification. Part number M28787/188-1 supersedes part number M28787/188. The two parts are interchangeable.

* 6.4 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life limits from table I.

Maximum power dissipation	
+5.5 V (7 mA)	38.5 mW
+16.5 V (20 mA)	330 mW
-16.5 V (-20 mA)	330 mW
Supply voltages	+5 V ±10 percent
	+15 V ±10 percent
	-15 V ±10 percent

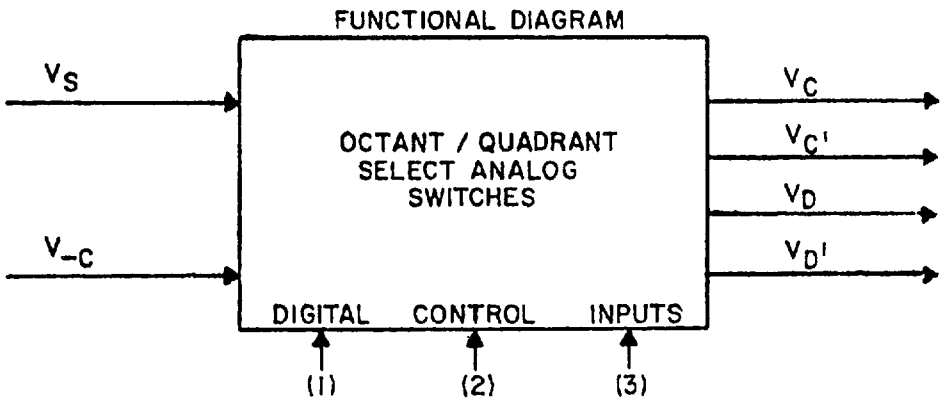
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Analog reference supply	
V output voltage (analog reference)	4.3 V dc minimum, 4.9 V dc maximum
2 V output voltage	5.8 V dc minimum, 6.6 V dc maximum
V output current (analog reference)	
Sink or source	1.4 mA maximum
2 V output current	
Sink or source	70 µA maximum
Octant/quadrant select	
S and -C input voltage	1.1 V rms maximum (biased to V)
S and -C impedance @ 400 Hz	1 MΩ minimum
C, C', D, and D' output voltage	1.1 V rms maximum (biased to V)
Angular accuracy	±0.03 degree
Phase shift	±0.5 degree
S and -C input protection	Reference test conditions B and C
Digital input voltage	
Low level	0.7 V maximum
High level	2 V minimum
Digital input current	
Low level (V_{IL} = 0 V)	-200 µA maximum
Low level (V_{IL} = 0.3 V)	-180 µA maximum
High level (V_{IH} = 2.4 V)	10 µA maximum
High level (V_{IH} = 5.5 V)	100 µA maximum

* 6.5 Module function. This module contains an analog reference voltage supply and select analog switches. The octant/quadrant detector selects octant or quadrant of the resolver shaft angle. Proper octant selection is controlled by the digital inputs which connect the resolver information to other circuits. The octant/quadrant module interfaces with a Scott-T transformer module and the MIL-M-28787/186 and MIL-M-28787/187 modules in digital/synchro and synchro/digital converter applications. The two analog signal inputs are buffered and have overvoltage protection.

6.5.1 Input to output relationship. The octant/quadrant select analog switches have the input to output relationship indicated below.

FUNCTIONAL DIAGRAM



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6.5.2 Octant selection transfer functions. The octant/quadrant select analog switches perform the following octant selection transfer functions.

Octant	Switch Input			Output Line			
	1	2	3	V_C	$V_{C'}$	V_D	$V_{D'}$
I	0	0	0	$\frac{\sqrt{2}}{2}[(V_S)+(V_{-C})]$	$+(V_S)$	$+(V_S)$	$+(V_{-C})$
II	0	0	1			$+(V_{-C})$	
III	0	1	0	$\frac{\sqrt{2}}{2}[-(V_S)+(V_{-C})]$	$-(V_S)$		
IV	0	1	1			$-(V_S)$	
V	1	0	0	$\frac{\sqrt{2}}{2}[-(V_S)-(V_{-C})]$			$-(V_{-C})$
VI	1	0	1			$-(V_{-C})$	
VII	1	1	0	$\frac{\sqrt{2}}{2}[(V_S)-(V_{-C})]$	$+(V_S)$		
VIII	1	1	1			$+(V_S)$	

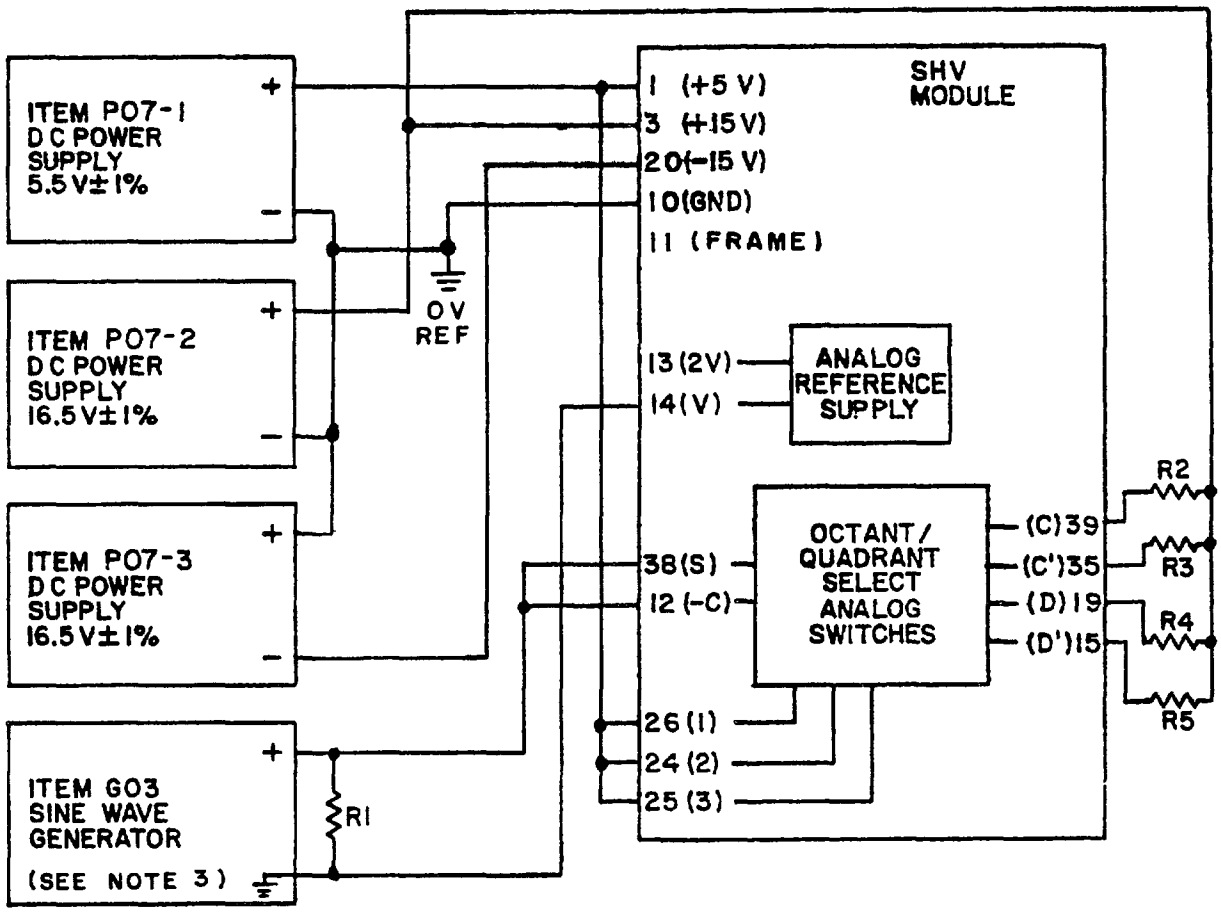
6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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NOTES:
1. R1 = 600Ω ±5%, 1 W.
2. R2 - R5 = 4.3 kΩ ±5%, 1/8 W.
3. Adjust item G03 for the following:
   Amplitude = 1.0 ±0.1 V rms.
   Frequency = 400 ±4 Hz.

FIGURE 2. Operating temperature and life test circuit.

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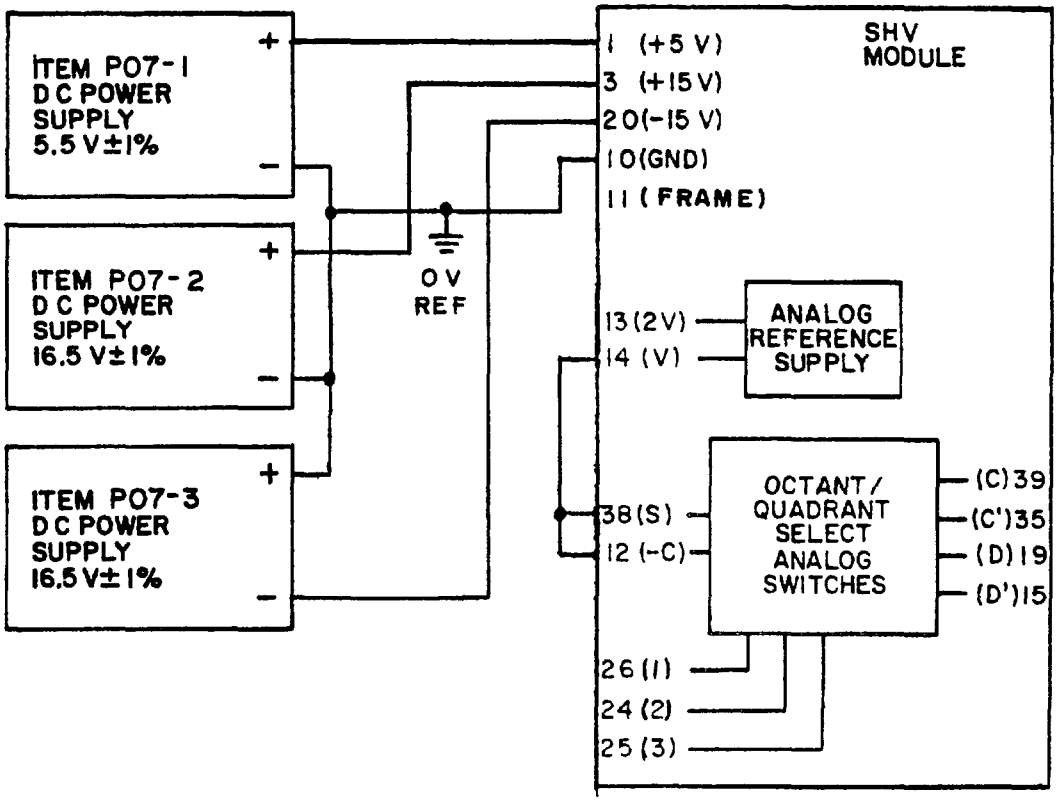


FIGURE 3. Component temperature test circuit.

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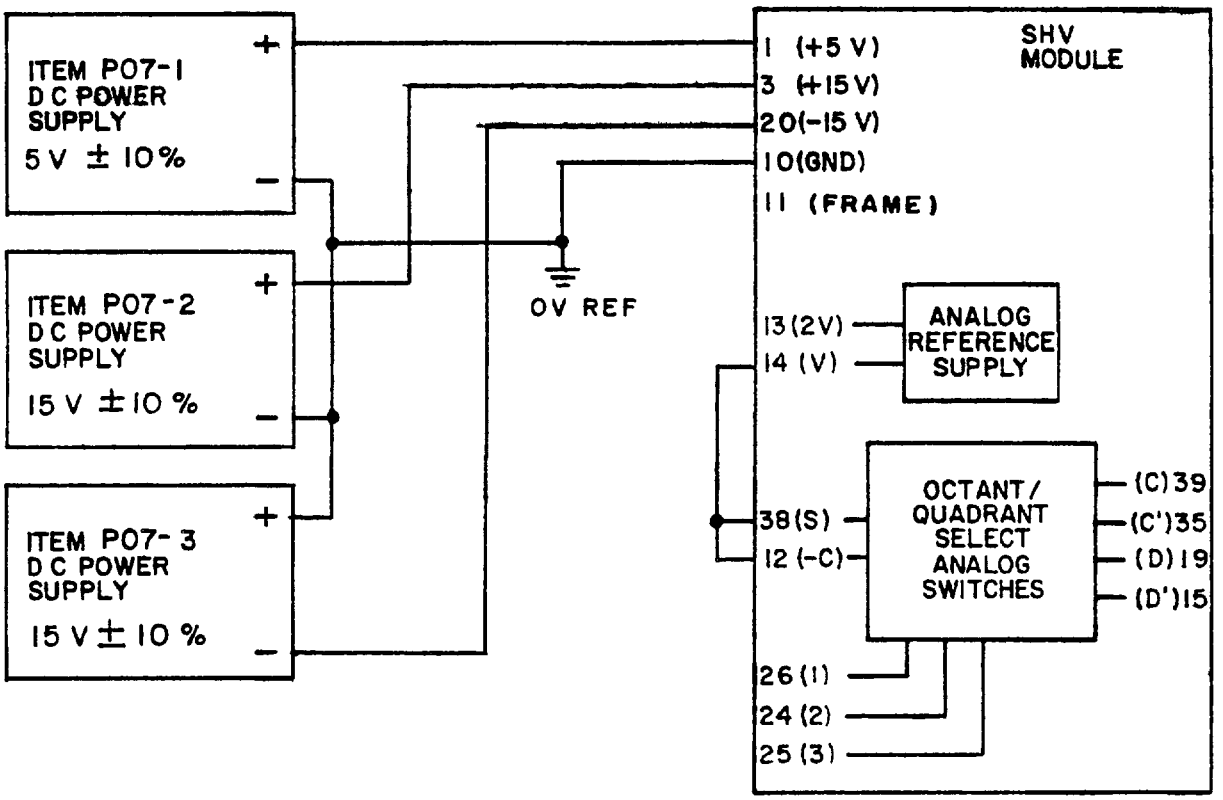


FIGURE 4. Vibration test circuit.

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TABLE II. Isolation test.

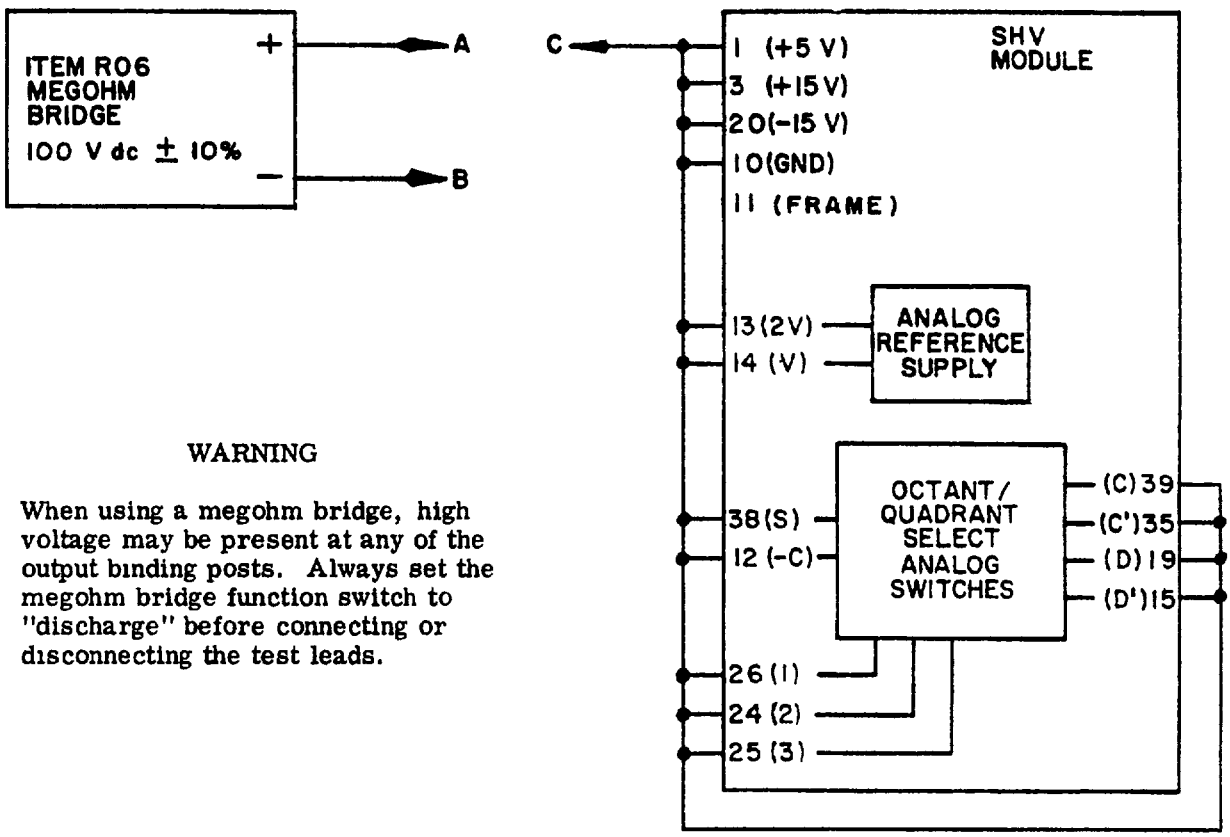
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation, frame	All pins except pin 11	11	10	∞	MΩ	5
A2	Isolation, used to unused pins	C	2,4,5,6,7,8,9,16,17,18,21,22, 23,27,28,29,30,31,32,33,34, 36,37,40	"	"	"	"
A3	Isolation, unused to unused pins	40	2,4,5,6,7,8,9,16,17,18,21,22, 23,27,28,29,30,31,32,33,34, 36,37	"	"	"	"
A4	"	37	2,4,5,6,7,8,9,16,17,18,21,22, 23,27,28,29,30,31,32,33,34, 36	"	"	"	"
A5	"	36	2,4,5,6,7,8,9,16,17,18,21,22, 23,27,28,29,30,31,32,33,34	"	"	"	"
A6	"	34	2,4,5,6,7,8,9,16,17,18,21,22, 23,27,28,29,30,31,32,33	"	"	"	"
A7	"	33	2,4,5,6,7,8,9,16,17,18,21,22, 23,27,28,29,30,31,32	"	"	"	"
A8	"	32	2,4,5,6,7,8,9,16,17,18,21,22, 23,27,28,29,30,31	"	"	"	"
A9	"	31	2,4,5,6,7,8,9,16,17,18,21,22, 23,27,28,29,30	"	"	"	"
A10	"	30	2,4,5,6,7,8,9,16,17,18,21,22, 23,27,28,29	"	"	"	"
A11	"	29	2,4,5,6,7,8,9,16,17,18,21,22, 23,27,28	"	"	"	"
A12	"	28	2,4,5,6,7,8,9,16,17,18,21,22, 23,27	"	"	"	"
A13	"	27	2,4,5,6,7,8,9,16,17,18,21,22, 23	"	"	"	"
A14	"	23	2,4,5,6,7,8,9,16,17,18,21,22	"	"	"	"

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TABLE II. Isolation test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A15	Isolation, unused to unused pins	22	2,4,5,6,7,8,9,16,17,18,21	10	∞	MΩ	5
A16	"	21	2,4,5,6,7,8,9,16,17,18	"	"	"	"
A17	"	18	2,4,5,6,7,8,9,16,17	"	"	"	"
A18	"	17	2,4,5,6,7,8,9,16	"	"	"	"
A19	"	16	2,4,5,6,7,8,9	"	"	"	"
A20	"	9	2,4,5,6,7,8	"	"	"	"
A21	"	8	2,4,5,6,7	"	"	"	"
A22	"	7	2,4,5,6	"	"	"	"
A23	"	6	2,4,5	"	"	"	"
A24	"	5	2,4	"	"	"	"
A25	"	4	2	"	"	"	"

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WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

FIGURE 5. Isolation test circuit for table II.

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TABLE III. Stress (overvoltage protection) check test.

Test condition	Parameter	Set 11CX4b	Initial limits 25°C ±5°C ambient		Unit	Test figure
			Min	Max		
B1	Stress (overvoltage protection) check	0°	- -	- -	- -	6
B2	Stress (overvoltage protection) check	90°	- -	- -	- -	6

TABLE IV. Overvoltage protection check test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
C	Overvoltage protection check	- -	- -	- -	7

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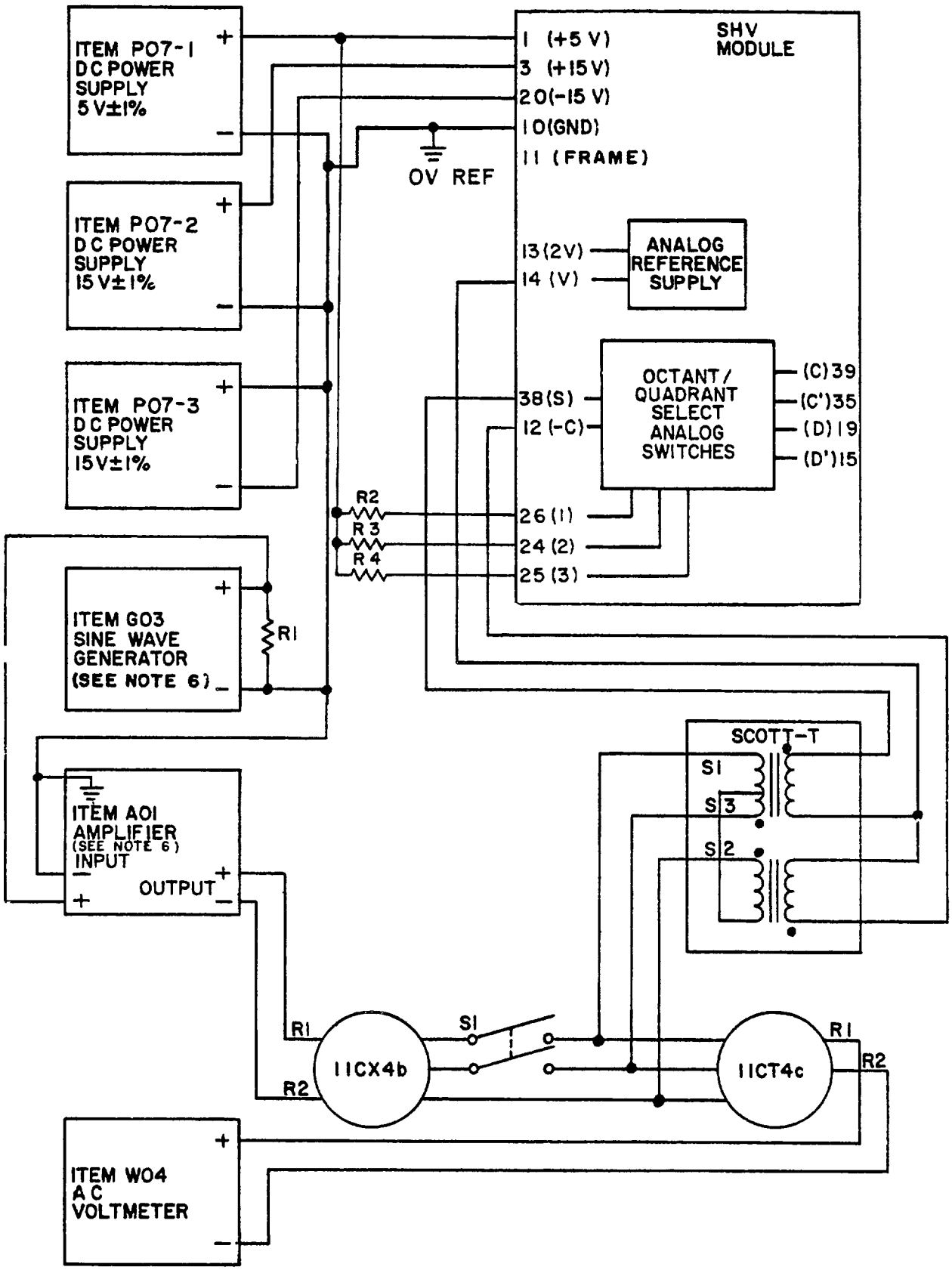


FIGURE 6. Stress (overvoltage protection) check test circuit for table III.

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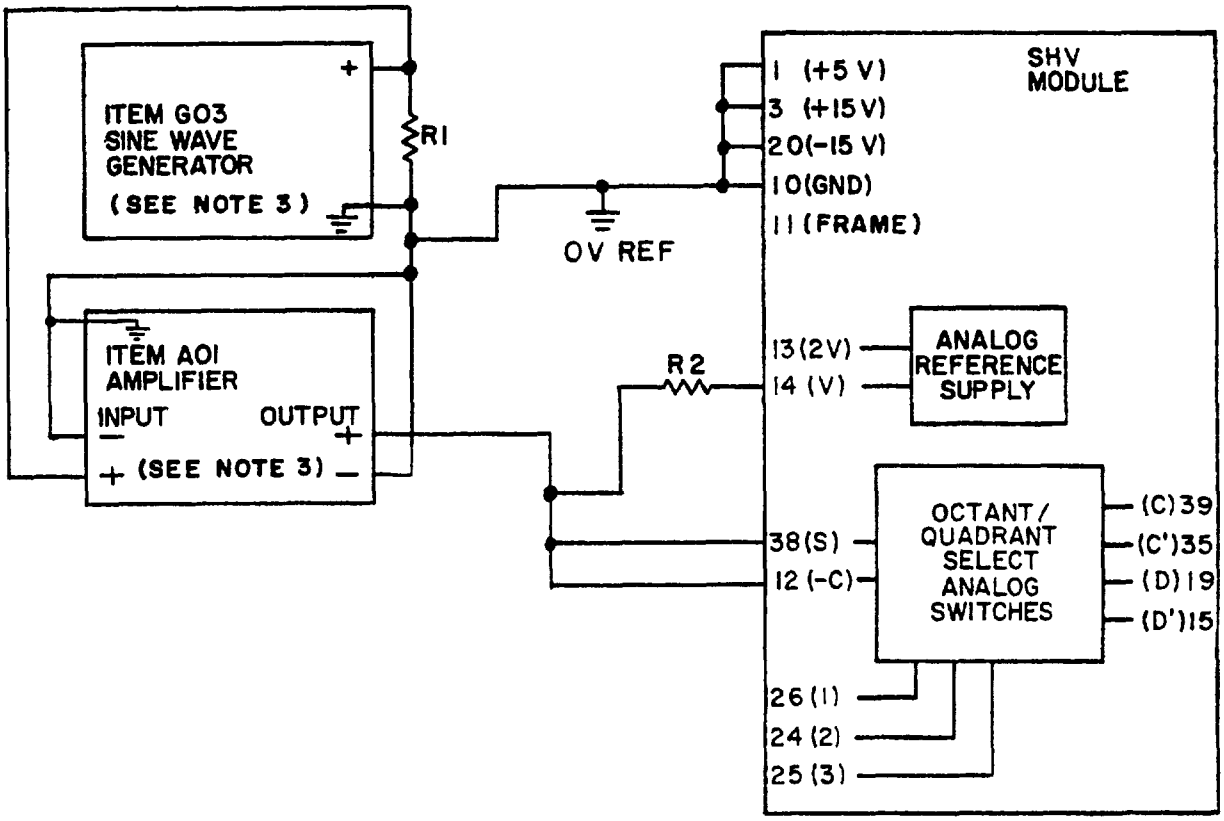
When using item A01 amplifier, high voltage may be present at the output terminals. Always set the voltage to zero before connecting or disconnecting the test leads. Resistor R1 must be across the input of item A01 at all times while power is applied to the amplifier.

NOTES:

1. R1 = 600Ω ±5%, 1 W.
2. R2-R4 = 2 kΩ ±5%, 1/8 W.
3. S1 = DPST switch.
4. Scott-T, Z_{SO} = 40 kΩ minimum matched to ±10%, 90 V L-L synchro signal in, 1 V rms ±1% resolver output.
5. 11CX4b is a synchro transmitter and 11CT4c is a synchro receiver.
6. Adjust items G03 and A01 for an output of 115 V rms ±5%, 400 ±4 Hz, at the output of item A01.
7. For test condition B1, set 11CX4b for 0 ±10 degrees. Adjust 11CT4c for a maximum voltage reading across R1 and R2. Open then close switch S1.
8. For test condition B2, set 11CX4b for 90 ±10 degrees. Adjust 11CT4c for a maximum voltage reading across R1 and R2. Open then close switch S1.
9. If the module passes the X-type tests, the module passes this test condition.

FIGURE 6. Stress (overvoltage protection) check test circuit for table III - Continued.

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WARNING

When using item A01 amplifier, high voltage may be present at the output terminals. Always set the voltage to zero before connecting or disconnecting the test leads.

Resistor R1 must be across the input of item A01 at all times while power is applied to the amplifier

NOTES:
1. R1 = 600Ω ±5%, 1 W.
2. R2 = 1 kΩ ±5%, 2 W.
3. Adjust items G03 and A01 for an output of 50 V_p ±5%, 400 ±20 Hz, at the output of item A01.
4. Run test for 5 to 10 seconds.
5. If the module passes the X-type tests, the module passes this test condition.

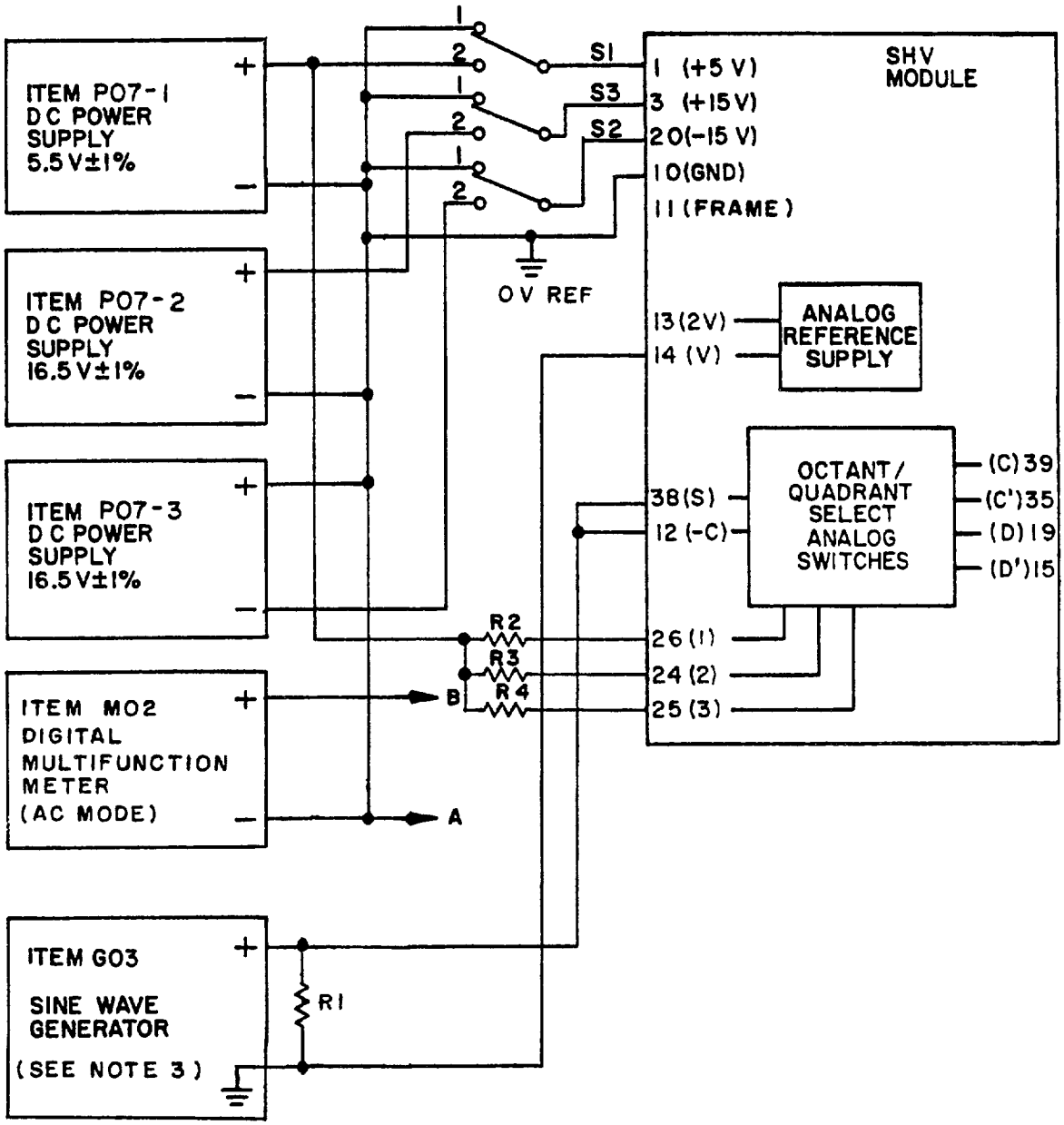
FIGURE 7. Overvoltage protection check test circuit for table IV.

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TABLE TABLE V. Hot socket test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D1	Hot socket	24	39	0.9	1.1	V rms	8
D2	"	Open	35	0.6	0.8	"	"
D3	"	"	19	"	"	"	"
D4	"	"	15	"	"	"	"
D5	"	26	15	"	"	"	"
D6	"	"	19	"	"	"	"
D7	"	"	35	"	"	"	"
D8	"	24, 26	39	0.9	1.1	"	"
D9	"	24, 26	39	0.9	1.1	"	"
D10	"	26	35	0.6	0.8	"	"
D11	"	"	19	"	"	"	"
D12	"	"	15	"	"	"	"
D13	"	Open	15	"	"	"	"
D14	"	"	19	"	"	"	"
D15	"	"	35	"	"	"	"
D16	"	24	39	0.9	1.1	"	"
D17	"	24	39	0.9	1.1	"	"
D18	"	Open	35	0.6	0.8	"	"
D19	"	"	19	"	"	"	"
D20	"	"	15	"	"	"	"
D21	"	26	15	"	"	"	"
D22	"	"	19	"	"	"	"
D23	"	"	35	"	"	"	"
D24	"	24, 26	39	0.9	1.1	"	"

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* FIGURE 8. Hot socket test circuit for table V.

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NOTES:
1. R1 = 600Ω ±5%, 1 W.
2. R2-R4 = 2 kΩ ±5%, 1/8 W.
3. S1-S3 = SPDT switch.
4. Item G03 setup:
   Amplitude = 0.707 V rms ±1%.
   Frequency = 400 Hz ±1%.
5. Switch sequencing:

   a. Initially set S1, S2, and S3 to position 1. Wait 10 seconds then switch each one to position 2 in the following order: S1, S2, then S3. Next taking readings for test conditions D1, D2, D3, D4, D5, D6, D7, and D8 in this order without any power disruption to the module.

   b. Reset S1, S2, and S3 to position 1. Wait 10 seconds then switch each one to position 2 in the following order: S2, S3, then S1. Next take readings for test conditions D9, D10, D11, D12, D13, D14, D15, and D16 in this order without any power disruption to the module.

   c. Reset S1, S2, and S3 to position 1. Wait 10 seconds then switch each one to position 2 in the following order: S3, S1, then S2. Next take readings for test conditions D17, D18, D19, D20, D21, D22, D23, and D24 in this order without any power disruption to the module.

* FIGURE 8. Hot socket test circuit for table V. Continued.

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TABLE VI. Input breakdown current (digital) test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
E1	1 Input breakdown current (digital)	26	0	100	µA	9
E2	2 "	24	"	"	"	"
E3	3 "	25	"	"	"	"

TABLE VII. Output short circuit current test.

Test condition	Parameter	Item F03 output angle	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
F1	C Output short circuit current	90°	39	10	50	mA	10
F2	C' "	"	35	"	"	"	"
F3	D "	"	19	"	"	"	"
F4	D' "	0°	15	"	"	"	"

TABLE VIII. V_S stress check test.

Test condition	Parameter	Item F03 output angle	Connect						Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	G to pin	Min	Max		
G	V_S stress check	45°	Open	Open	Open	Open	Open	Open	---	---	---	11

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TABLE IX. Angular accuracy check and quadrature test.

Test condition	Parameter	Item F03 output angle	Connect						Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	G to pin	Min	Max		
H1	Angular accuracy check	0°	Open	24,25, 26	19,K	39,L	Open	Open	359.970	0.030	Degrees	11
H2	Quadrature	0°	"	"	"	"	"	"	-2	2	mV	"
H3	Angular accuracy check	22.5°	"	"	"	"	"	"	22.470	22.530	Degrees	"
H4	Quadrature	22.5°	"	"	"	"	"	"	-2	2	mV	"
H5	Angular accuracy check	45°	25	24,26	"	"	"	"	35.234	35.294	Degrees	"
H6	Quadrature	45°	"	"	"	"	"	"	-2	2	mV	"
H7	Angular accuracy check	67.5°	"	"	"	"	"	"	22.470	22.530	Degrees	"
H8	Quadrature	67.5°	"	"	"	"	"	"	-2	2	mV	"
H9	Angular accuracy check	90°	24	25,26	"	"	"	"	179.970	180.030	Degrees	"
H10	Quadrature	90°	24	25,26	"	"	"	"	-2	2	mV	"
H11	Angular accuracy check	22.5°	Open	24,25, 26	15,H	35,J	"	14	67.470	67.530	Degrees	"
H12	Quadrature	"	Open	24,25, 26	"	"	"	"	-2	2	mV	"
H13	Angular accuracy check	"	24	25,26	"	"	"	"	112.470	112.530	Degrees	"
H14	Quadrature	"	24	25,26	"	"	"	"	-2	2	mV	"
H15	Angular accuracy check	"	26	24,25	"	"	"	"	247.470	247.530	Degrees	"
H16	Quadrature	"	26	24,25	"	"	"	"	-2	2	mV	"
H17	Angular accuracy check	"	24,26	25	"	"	"	"	292.470	292.530	Degrees	"
H18	Quadrature	"	24,26	25	"	"	"	"	-2	2	mV	"
H19	Angular accuracy check	"	24,25	26	19,K	39,L	"	Open	314.970	315.030	Degrees	"
H20	"	"	26	24,25	"	"	"	"	202.470	202.530	"	"
H21	"	"	25,26	24	"	"	"	"	224.970	225.030	"	"

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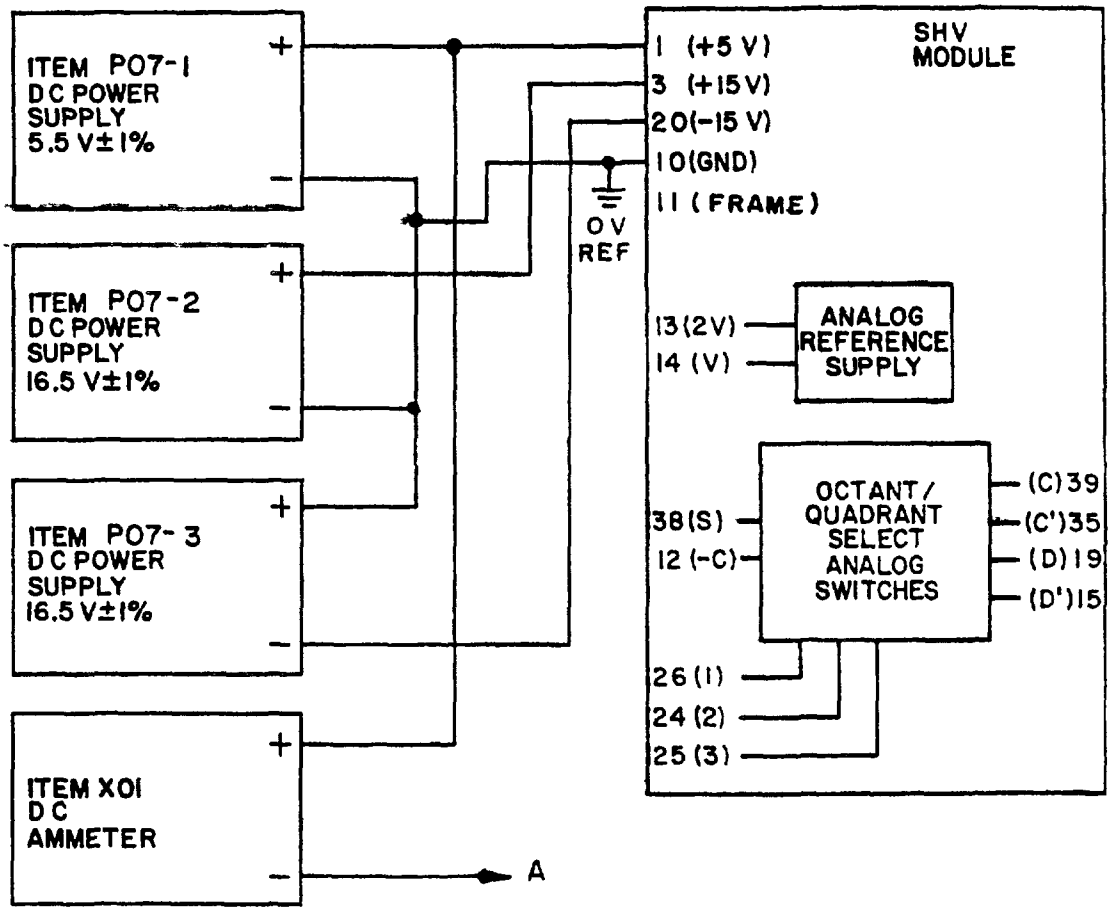
TABLE IX. Angular accuracy check and quadrature test - Continued.

Test condition	Parameter	Item F03 output angle	Connect						Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	G to pin	Min	Max		
H22	Angular accuracy check	22.5°	24,26	25	19,K	39,L	Open	Open	247.470	247.530	Degrees	11
H23	"	22.5°	24,25, 26	Open	"	"	"	"	134.970	135.030	"	"
H24	"	315°	Open	24,25, 26	"	"	"	"	269.970	270.030	"	"
H25	"	45°	24,25, 26	Open	"	"	"	"	89.970	90.030	"	"
H26	"	0°	25	24,26	"	"	"	"	54.7056	54.7656	"	"
H27	"	90°	26	24,25	"	"	"	"	234.7056	234.7656	"	"

TABLE X. Absolute Tr and phase shift (quadrature) test.

Test condition	Parameter	Item F03 output angle	Connect						Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	G to pin	Min	Max		
J1	Absolute Tr	90°	Open	24,25, 26	F	35	Open	Open	44.7121	45.2851	Degrees	11
J2	Phase shift (quadrature)	"	"	"	Open	Open	35	"	-0.5	0.5	"	"
J3	Absolute Tr	"	"	"	F	19	Open	"	44.7121	45.2851	"	"
J4	Phase shift (quadrature)	"	"	"	Open	Open	19	"	-0.5	0.5	"	"
J5	Absolute Tr	"	"	"	38	35	Open	"	J1- 0.040	J1+ 0.001	"	"
J6	Absolute Tr	"	"	"	38	19	Open	"	J3- 0.040	J3+ 0.001	"	"

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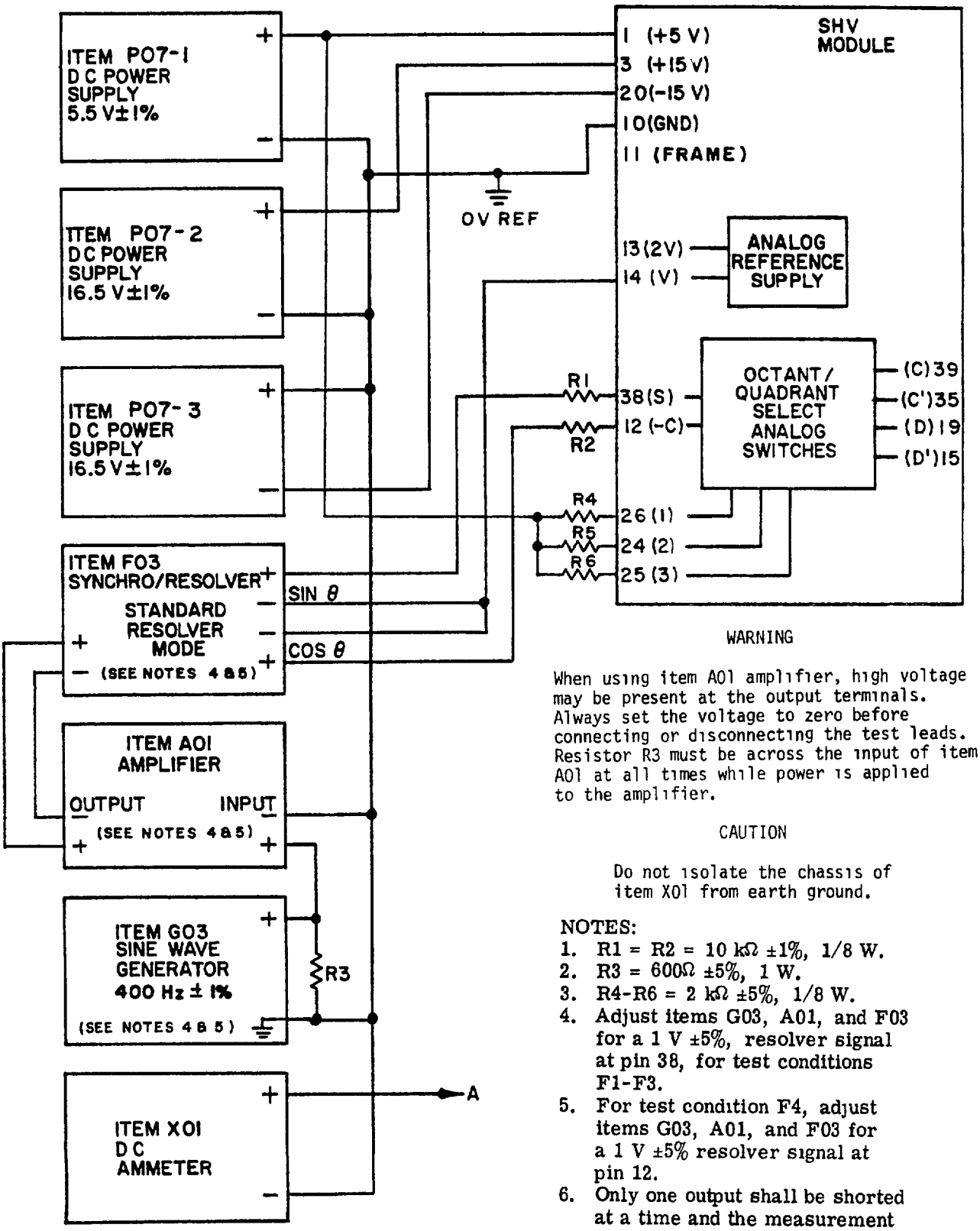


CAUTION

Do not isolate the chassis of item XO1 from earth ground.

FIGURE 9. Input breakdown current (digital) test circuit for table VI.

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WARNING

When using item A01 amplifier, high voltage may be present at the output terminals. Always set the voltage to zero before connecting or disconnecting the test leads. Resistor R3 must be across the input of item A01 at all times while power is applied to the amplifier.

CAUTION

Do not isolate the chassis of item X01 from earth ground.

NOTES:

1. R1 = R2 = 10 kΩ ±1%, 1/8 W.
2. R3 = 600Ω ±5%, 1 W.
3. R4-R6 = 2 kΩ ±5%, 1/8 W.
4. Adjust items G03, A01, and F03 for a 1 V ±5%, resolver signal at pin 38, for test conditions F1-F3.
5. For test condition F4, adjust items G03, A01, and F03 for a 1 V ±5% resolver signal at pin 12.
6. Only one output shall be shorted at a time and the measurement taken within 10 seconds.

FIGURE 10. Output short circuit current test circuit for table VII.

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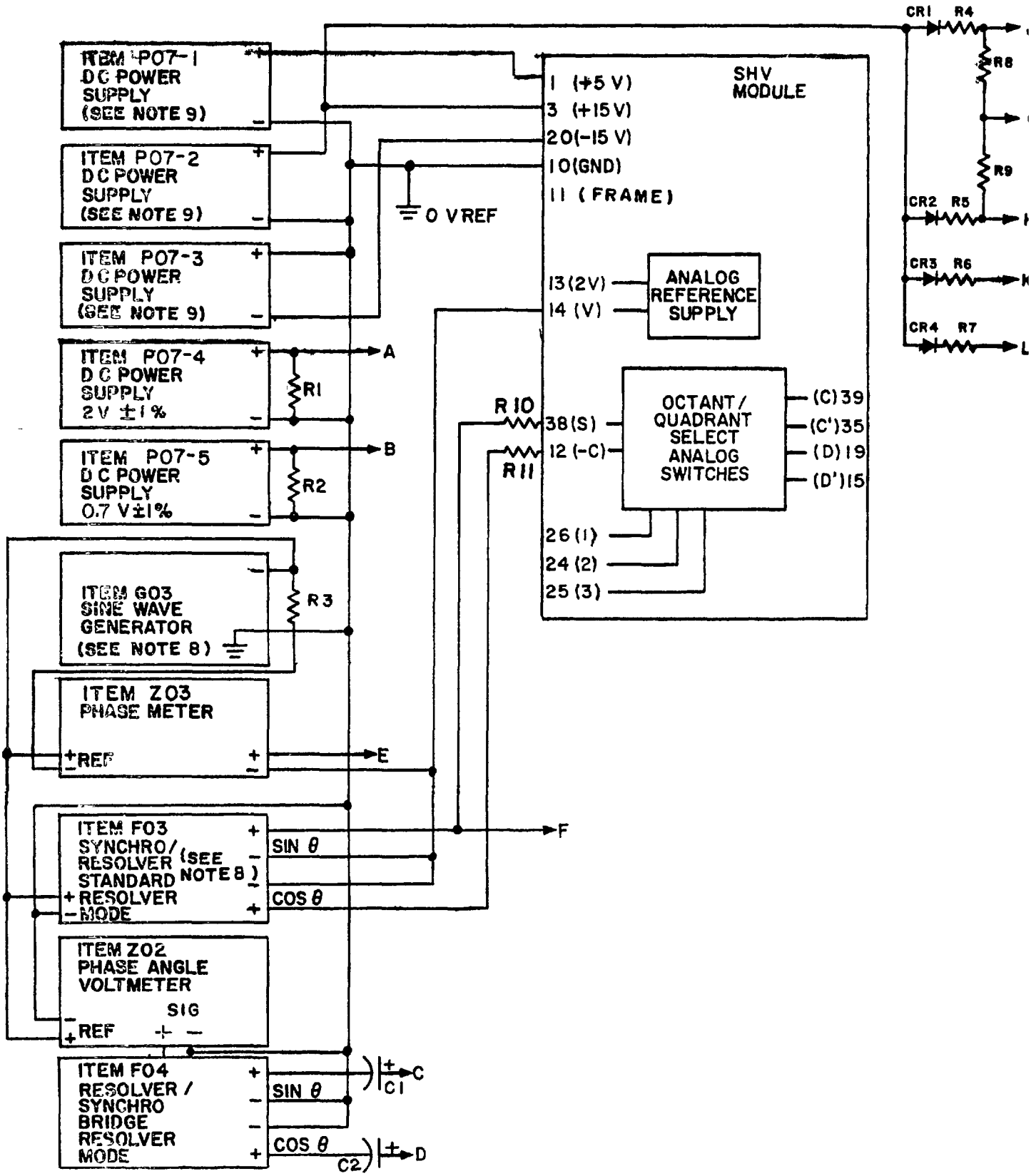


FIGURE 11. V_S stress and angular accuracy check, quadrature, absolute Tr, and phase shift (quadrature) test circuit for tables VIII, IX, and X.

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NOTES:

1. R1 = R2 = 50Ω ±5%, 1/4 W.
2. R3 = 600Ω ±5Ω, 1 W.
3. R4-R7 = Select values for a current flow of 5.6 mA ±5% through each resistor when points J, K, H, and L are held at 4.6 V dc.
4. R8 = R9 = 77Ω ±1%, 1/8 W.
5. R10 = R11 = 2 kΩ ±1%, 1/8 W.
6. C1 = C2 = 470 µF ±10%, 20 WV dc, electrolytic.
7. CR1-CR4 = 1N916, or equivalent.
8. For test conditions G, H, and J, adjust items G03 and F03 for a resolver signal output of 1 V rms ±5%, 400 ±10 Hz, out of item F03.
9. Adjust items P07-1, P07-2, and P07-3 as follows:

Test condition	P07-1	P07-2	P07-3
G	7 V ±1%	18 V ±1%	18 V ±1%
H1-H27	5 V ±1%	15 V ±1%	15 V ±1%
J1-J6	5 V ±1%	15 V ±1%	15 V ±1%

Never decrease the voltage to pin 1 below 4.5 V dc when doing test conditions G, H, and J. Never decrease the voltage to pin 3 below 13.5 V dc when doing test conditions G, H, and J. Never increase the voltage to pin 20 above -13.5 V dc when doing test conditions G, H, and J.

10. Make all ground connections at a common point when making circuit connections.
11. Before doing quadrature tests: Set item F03 to 0 degrees; connect C to +sinθ output of item F03 and D to +cosθ output of item F03; set item F04 to 90 degrees; take voltage reading on item Z02, let reading equal M, then multiply quadrature readings by 1/M before comparing number with test condition limits.

FIGURE 11. V_S stress and angular accuracy check, quadrature, absolute Tr, and phase shift (quadrature) test circuit for tables VIII, IX, and X - Continued.

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TABLE XI. Power supply current test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
K1	Power supply current (+5.5 V supply)	0	7	mA	12
K2	Power supply current (+16.5 V supply)	"	20	"	"
K3	Power supply current (-16.5 V supply)	"	20	"	"

TABLE XII. Reference supply output voltage and reference supply output short circuit current test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
L1	2 V Reference supply output voltage (ac)	Open	Open	13	0	10	mV rms	13
L2	2 V Reference supply output voltage (dc)	13	"	Open	5.8	6.6	V	"
L3	V Reference supply output voltage (ac, analog ground)	Open	"	14	0	10	mV rms	"
L4	V Reference supply output voltage (dc)	14	Open	Open	4.3	4.9	V	"
M1	2 V Reference supply output short circuit current	Open	13	"	1.5	25.0	mA	"
M2	V Reference supply output short circuit current (analog ground)	Oper.	14	"	10	30	mA	"

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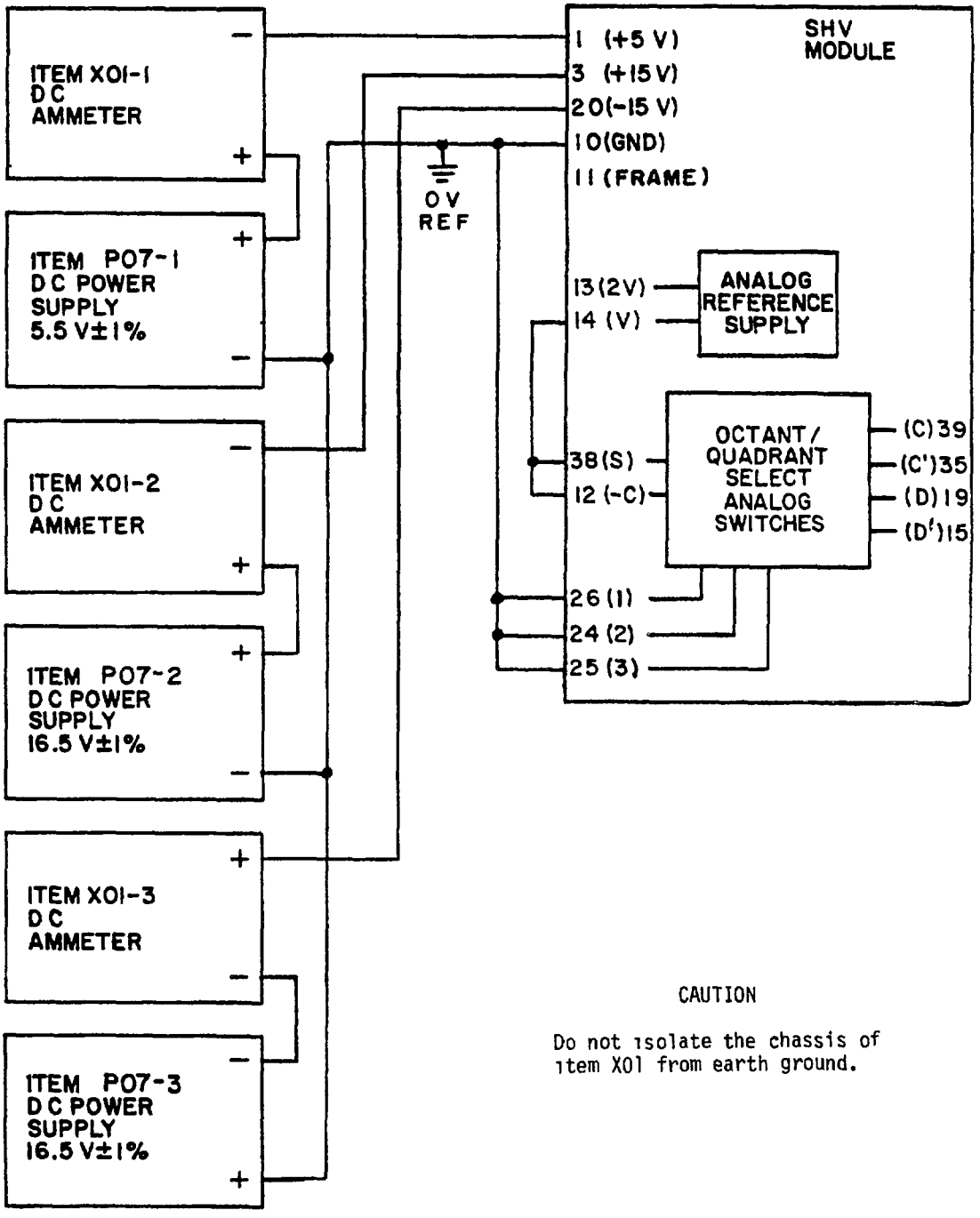


FIGURE 12. Power supply current test circuit for table XI.

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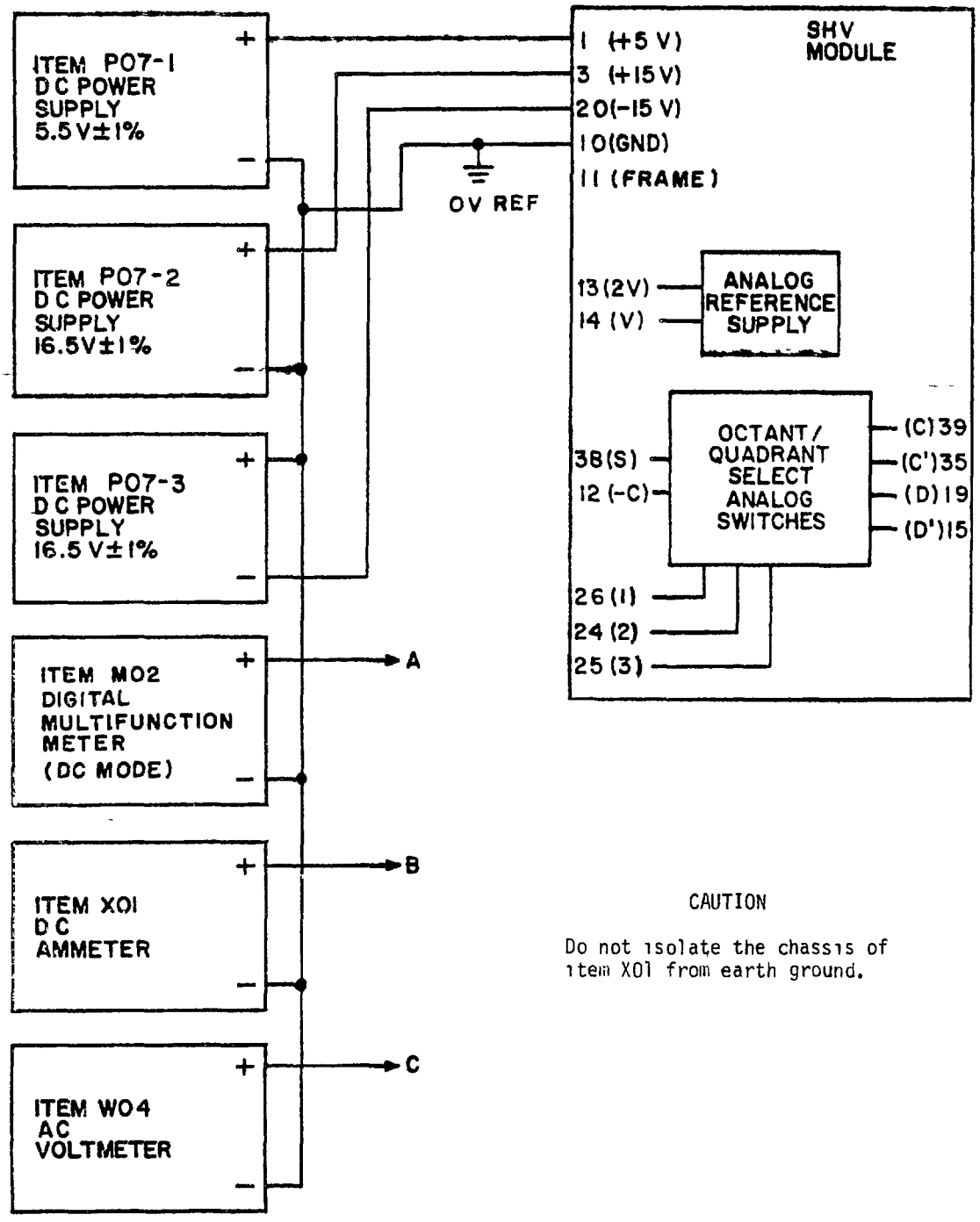
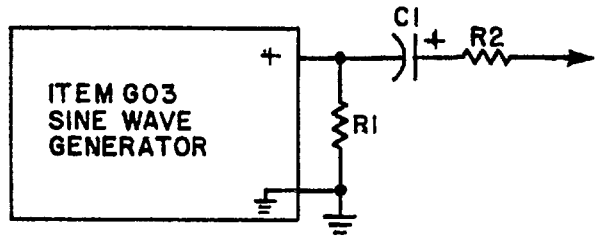


FIGURE 13. Reference supply output voltage and reference supply output short circuit current test circuit for table XII.

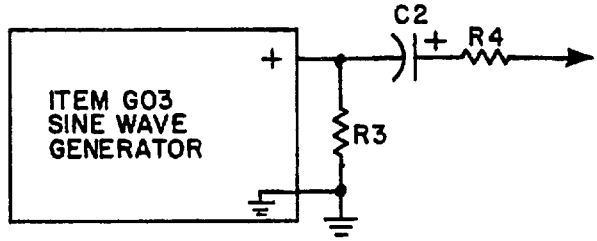
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NOTES:
1. With point B disconnected, item X01 should indicate zero current on the 1 µA range.
2. When performing the output short circuit current test, only one output shall be shorted at a time and the measurement taken within 5 seconds.
3. Load for test conditions L1 and L2. Connect to pin 13.



(a) R1 = 600Ω ±5%, 1 W.
(b) R2 = 20 kΩ ±1%, 1/8 W.
(c) C1 = 1,000 µF ±10%, 20 WV dc, electrolytic.
(d) Set output of item G03 for 1 V rms ±1%, 400 ±5 Hz.

4. Load for test conditions L3 and L4. Connect to pin 14.



(a) R3 = 600Ω ±5%, 1 W.
(b) R4 = 1.0 kΩ ±1%, 1/8 W.
(c) C2 = 1,000 µF ±10%, 20 WV dc, electrolytic.
(d) Set output of item G03 for 1 V rms ±1%, 400 ±5 Hz.

5. The loads specified in notes 3 and 4 cannot be connected at the same time.

FIGURE 13. Reference supply output voltage and reference supply output short circuit current test circuit for table XII - Continued.

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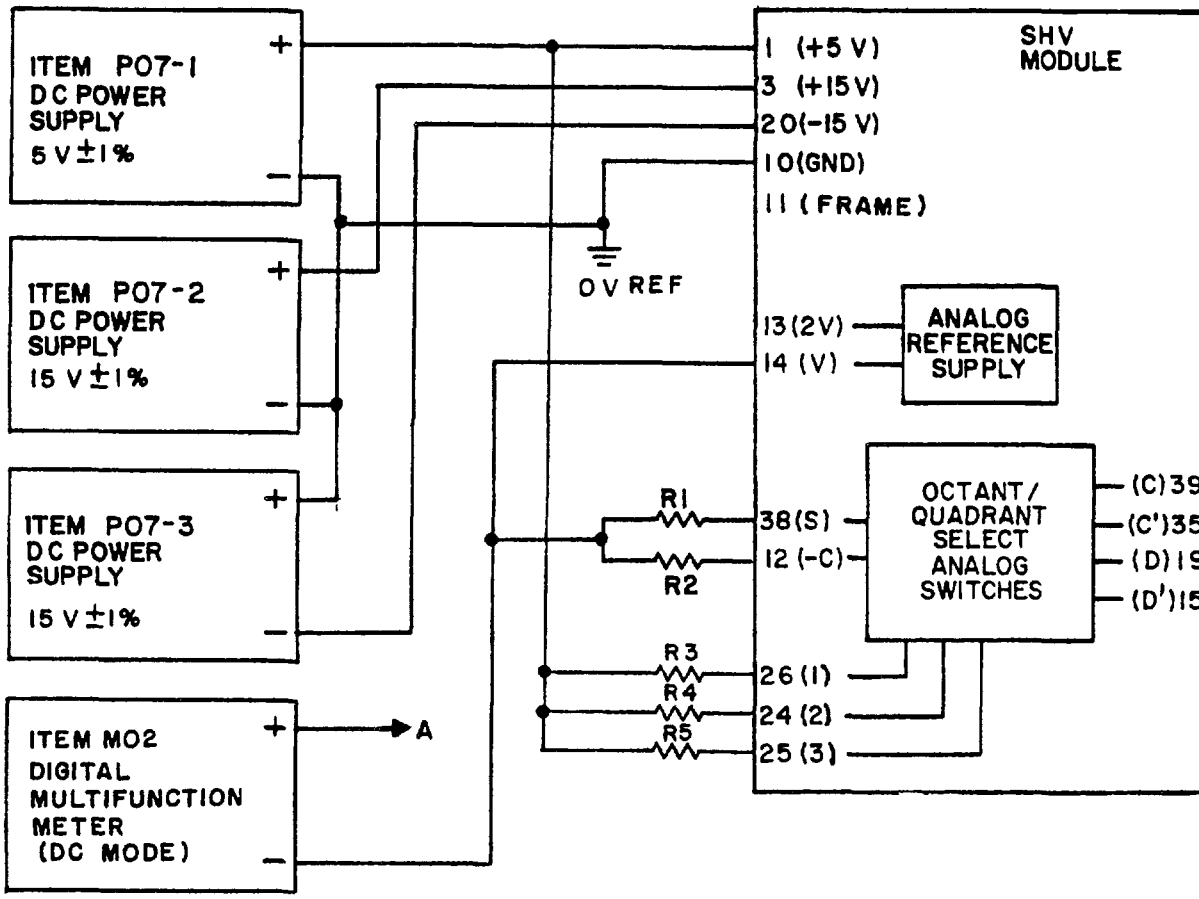
TABLE XIII. Output offset voltage (analog) test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
N1	Output offset voltage (analog)	39	-25	25	mV	14
N2	"	35	"	"	"	"
N3	"	19	"	"	"	"
N4	"	15	"	"	"	"

TABLE XIV. Input current (digital) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
P1	2 Input current (digital), high level	Open	24	0	10	µA	15
P2	1 "	"	26	"	"	"	"
P3	3 "	"	25	"	"	"	"
P4	2 Input current (digital), low level	24	Open	-10	-200	"	"
P5	1 "	26	"	"	"	"	"
P6	3 "	25	"	"	"	"	"

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NOTES:
1. R1 = R2 = 100Ω ±1%, 1/8 W.
2. R3-R5 = 2 kΩ ±1%, 1/8 W

FIGURE 14. Output offset voltage (analog) test circuit for table XIII.

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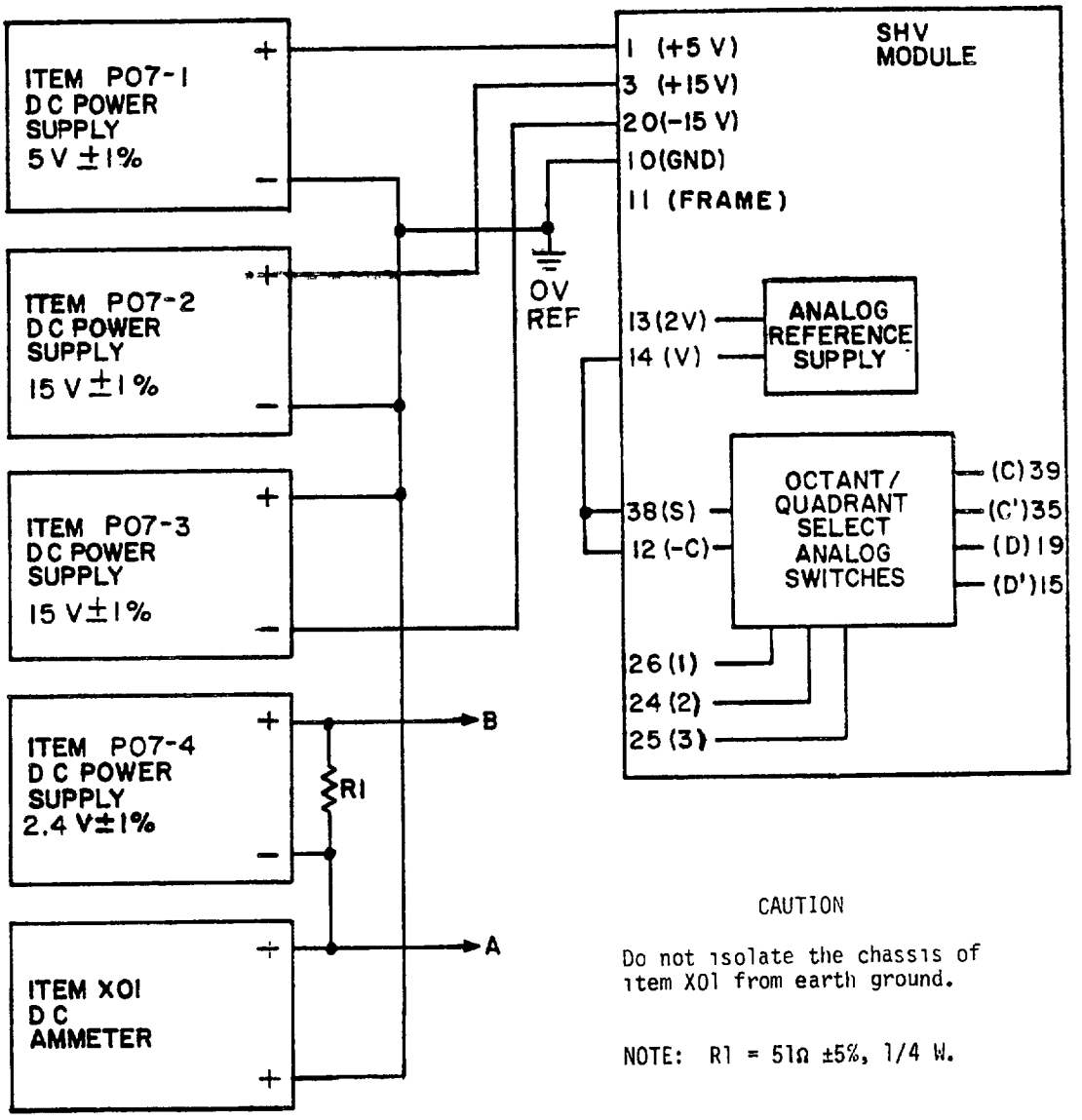


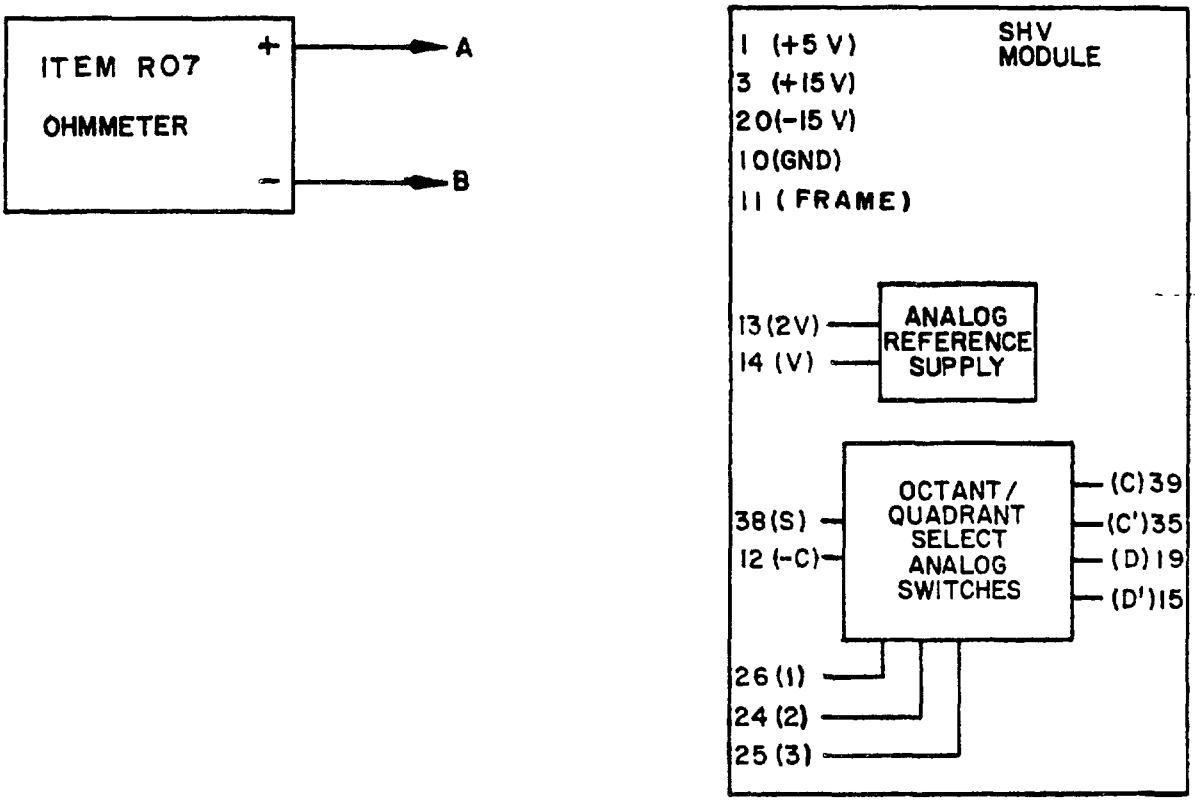
FIGURE 15. Input current (digital) test circuit for table XIV.

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TABLE XV. Continuity test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
Q	Continuity, frame	11	Frame	0	1	Ω	16

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NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

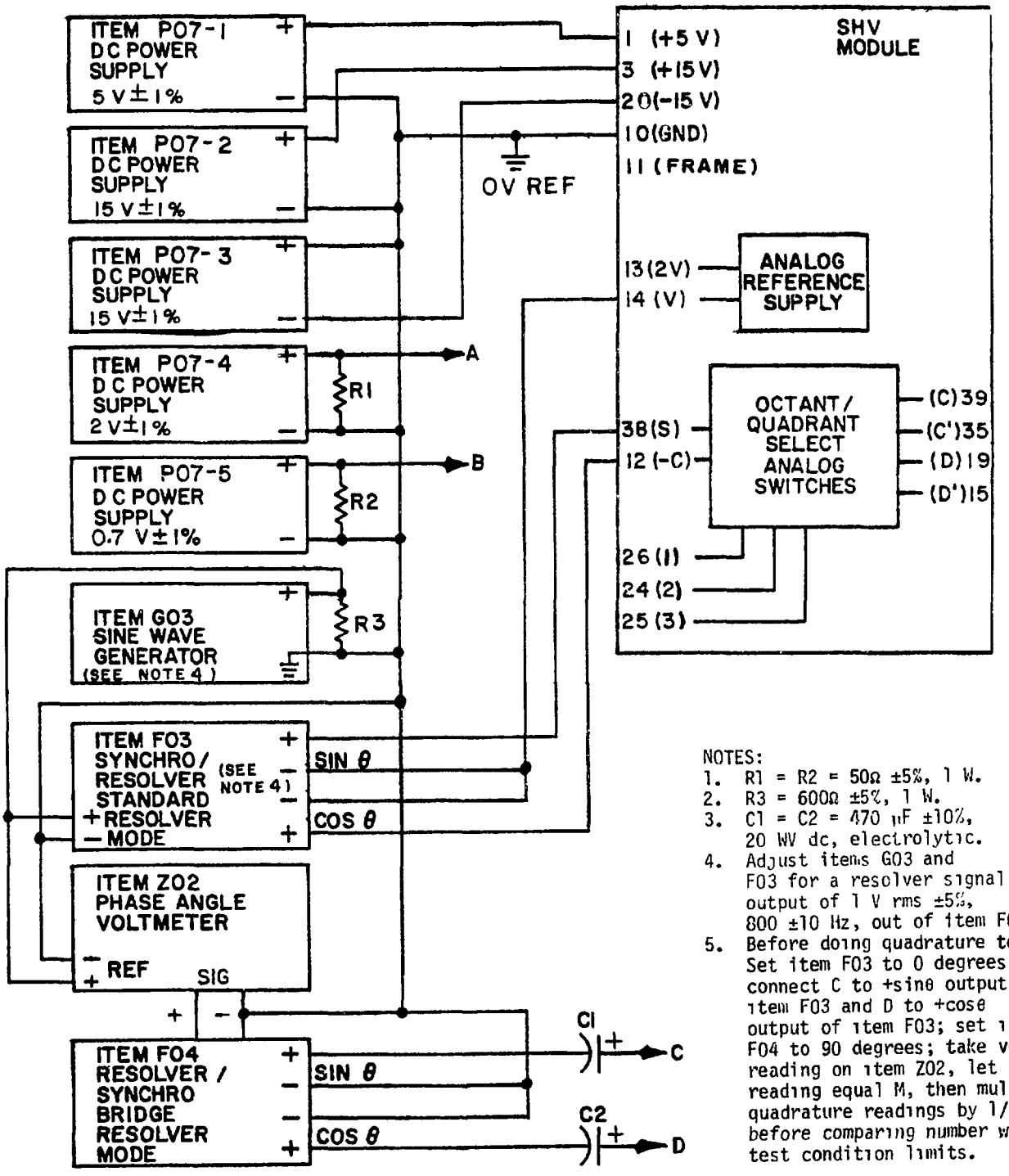
* FIGURE 16. Continuity test circuit for table XV.

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TABLE XVI. Angular accuracy (800 Hz) check and quadrature (800 Hz) test.

Test condition	Parameter	Item F03 output angle	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R1	Angular accuracy (800 Hz) check	45°	Open	24,25,26	19	39	35.234	35.394	Degrees	17
R2	Quadrature (800 Hz)	"	Open	24,25,26	"	"	-4	4	mV	"
R3	Angular accuracy (800 Hz) check	"	25	24,26	"	"	35.234	35.394	Degrees	"
R4	Quadrature (800 Hz)	"	25	24,26	"	"	-4	4	mV	"
R5	Angular accuracy (800 Hz) check	"	24	25,26	"	"	89.970	90.030	Degrees	"
R6	Quadrature (800 Hz)	"	24	25,26	"	"	-4	4	mV	"
R7	Angular accuracy (800 Hz) check	"	26	24,25	"	"	215.234	215.294	Degrees	"
R8	Quadrature (800 Hz)	"	26	24,25	"	"	-4	4	mV	"
R9	Angular accuracy (800 Hz) check	"	24,26	25	"	"	269.970	270.030	Degrees	"
R10	Quadrature (800 Hz)	"	24,26	25	"	"	-4	4	mV	"
R11	Angular accuracy (800 Hz) check	"	Open	24,25,26	15	35	44.970	45.030	Degrees	"
R12	Quadrature (800 Hz)	"	Open	24,25,26	"	"	-4	4	mV	"
R13	Angular accuracy (800 Hz) check	"	24	25,26	"	"	134.970	135.030	Degrees	"
R14	Quadrature (800 Hz)	"	24	25,26	"	"	-4	4	mV	"
R15	Angular accuracy (800 Hz) check	"	26	24,25	"	"	224.970	225.030	Degrees	"
R16	Quadrature (800 Hz)	"	26	24,25	"	"	-4	4	mV	"
R17	Angular accuracy (800 Hz) check	"	24,26	25	"	"	314.970	315.030	Degrees	"
R18	Quadrature (800 Hz)	"	24,26	25	"	"	-4	4	mV	"

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NOTES:
1. R1 = R2 = 50Ω ±5%, 1 W.
2. R3 = 600Ω ±5%, 1 W.
3. C1 = C2 = 470 μF ±10%, 20 WV dc, electrolytic.
4. Adjust items G03 and F03 for a resolver signal output of 1 V rms ±5%, 800 ±10 Hz, out of item F03.
5. Before doing quadrature tests: Set item F03 to 0 degrees; connect C to +sinθ output of item F03 and D to +cosθ output of item F03; set item F04 to 90 degrees; take voltage reading on item Z02, let reading equal M, then multiply quadrature readings by 1/M before comparing number with test condition limits.

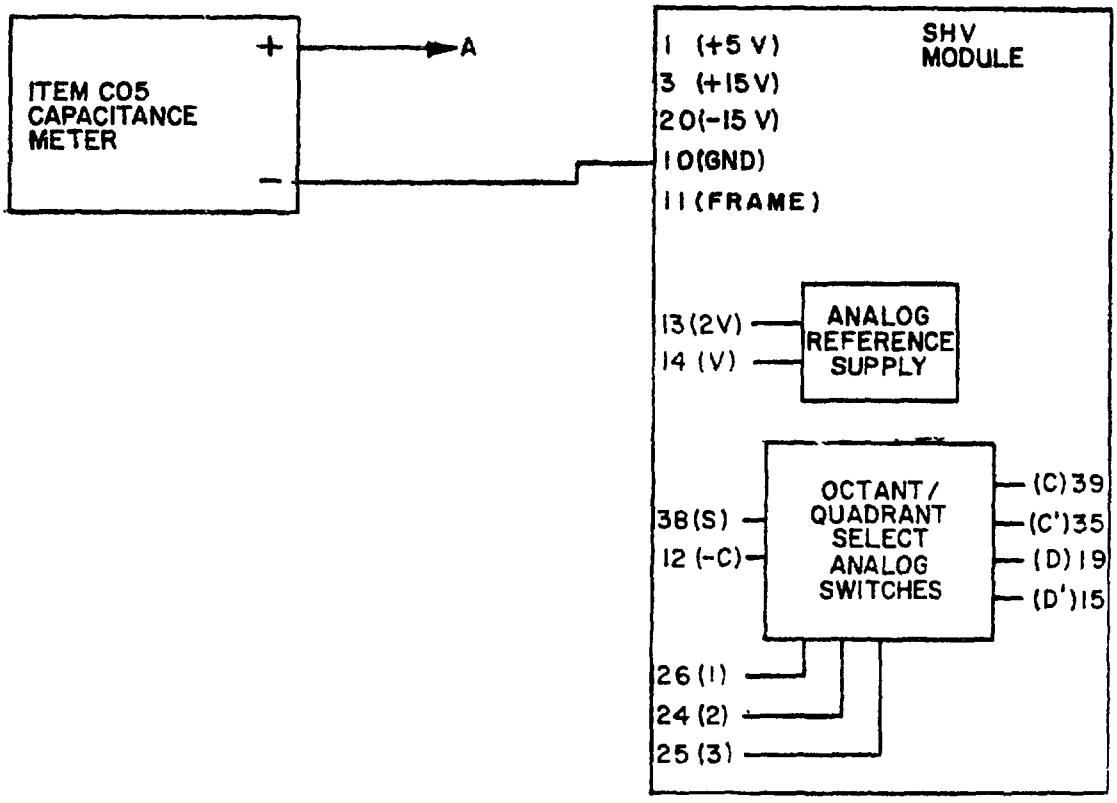
FIGURE 17. Angular accuracy (800 Hz) check and quadrature (800 Hz) test circuit for table XVI.

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TABLE XVII. Input capacitance (digital) test.

Test condition	Parameter,	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
S1	2 Input capacitance (digital)	24	2	15	pF	18
S2	3 "	25	"	"	"	"
S3	1 "	26	"	"	"	"

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NOTES.
1. Null out the effect of the test leads prior to connecting to the module.
2. Do not attach test connectors or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may effect the readings.

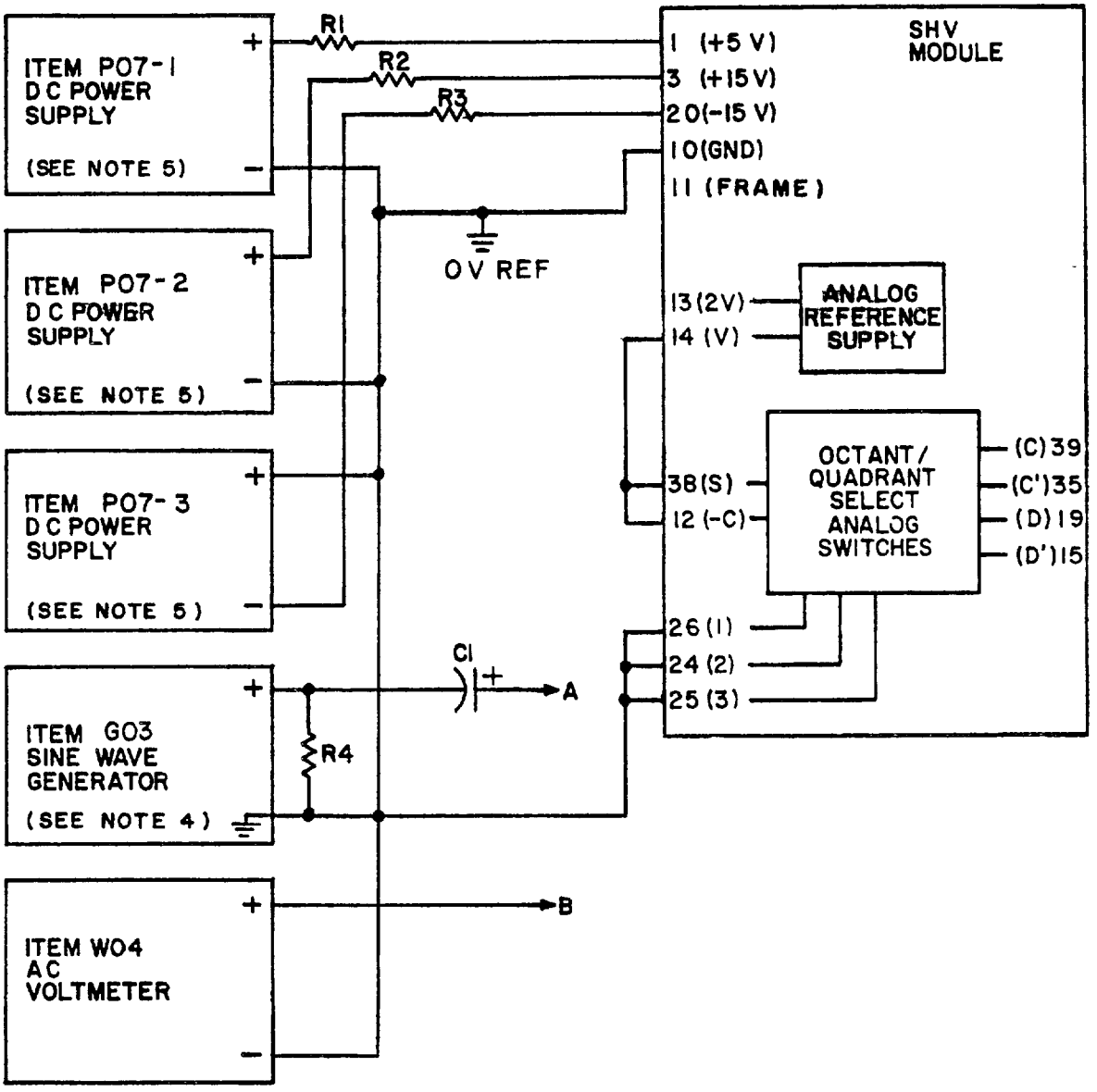
FIGURE 18. Input capacitance (digital) test circuit for table XVII.

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TABLE XVIII. Noise rejection test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
T1	Noise rejection	1	39	0.0	3.5	mV rms	19
T2	"	"	35	"	"	"	"
T3	"	"	19	"	"	"	"
T4	"	"	15	"	"	"	"
T5	"	3	39	"	"	"	"
T6	"	"	35	"	"	"	"
T7	"	"	19	"	"	"	"
T8	"	"	15	"	"	"	"
T9	"	20	39	"	"	"	"
T10	"	"	35	"	"	"	"
T11	"	"	19	"	"	"	"
T12	"	"	15	"	"	"	"
T13	Calculation T1 + T5 + T9	--	--	"	"	"	"
T14	Calculation T2 + T6 + T10	--	--	"	"	"	"
T15	Calculation T3 + T7 + T11	--	--	"	"	"	"
T16	Calculation T4 + T8 + T12	--	--	"	"	"	"

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NOTES.
1. R1-R3 = 100Ω ±0.5%, 1/8 W.
2. R4 = 600Ω ±5%, 1 W.
3. C1 = 470 µF ±10%, 20 WV dc, electrolytic.
4. After point A has been connected adjust item G03 for 1.00 ±0.01 V rms, 400 ±10 Hz, at point A.
5. Adjust item P01-7 for 5 V ±1% at module pin 1; adjust item P07-2 for 15 V ±1% at module pin 3; and adjust item P07-3 for -15 V ±1% at module pin 20.
6. For test conditions T9-T12, reverse the polarity of C1.

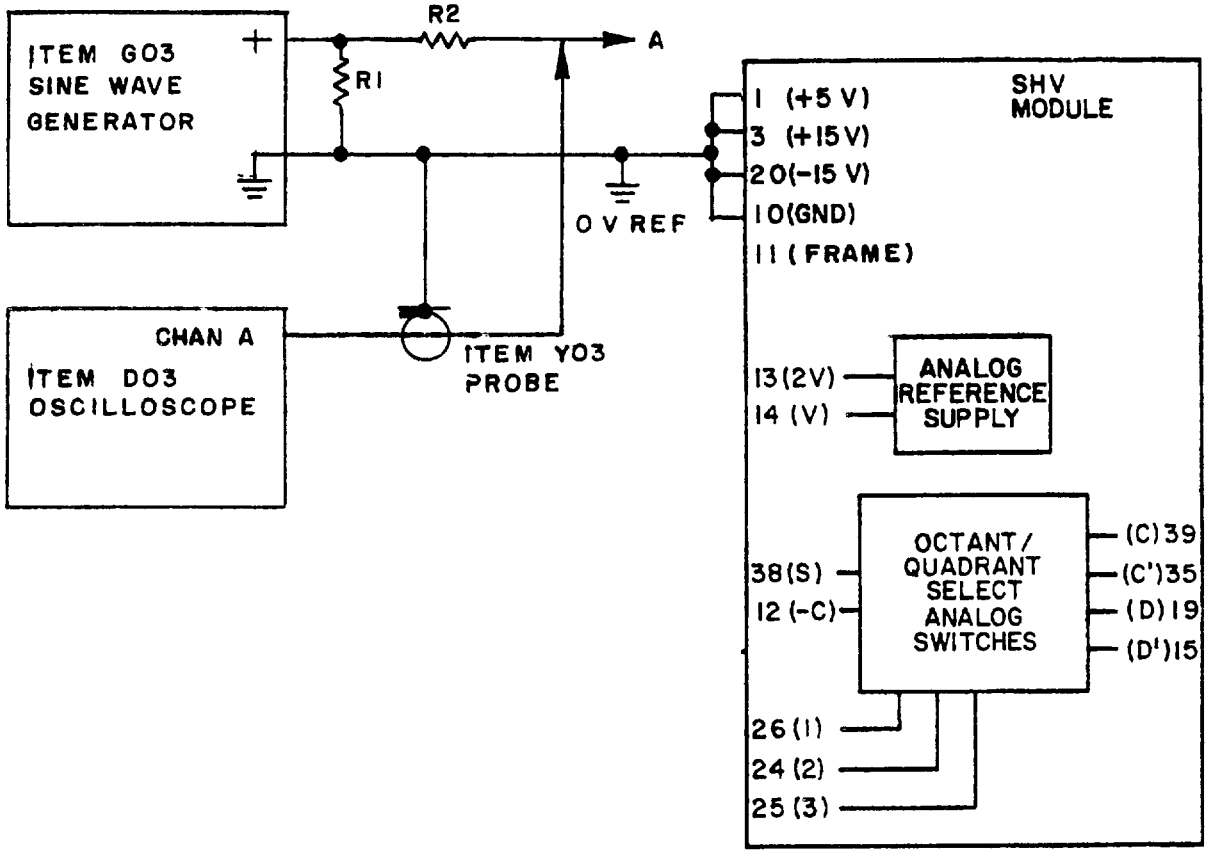
FIGURE 19. Noise rejection test circuit for table XVIII.

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TABLE XIX. Input impedance (power off) check.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
U1	S Input impedance (power off)	38	0	240	mVp	20
U2	-C Input impedance (power off)	12	0	240	mVp	20

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NOTES.
1. R1 = 600Ω ±5%, 1 W.
2. R2 = 2 kΩ ±1%, 1/8 W.
3. Item GO3 setup:
   Amplitude = 1.100 ±0.010 V rms.
   Frequency = 400 ±10 Hz.
4. This test verifies an input impedance of 10 kΩ, minimum.
5. Measure the largest peak excursion from the zero volt reference.
6. Adjust item GO3 with point A connected.

FIGURE 20. Input impedance (power off) check test circuit for table XIX.

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